

Clinical Criteria for Systemic Lupus Erythematosus (SLE)

The classification of systemic lupus erythematosus is based on 11 criteria. A person is considered to have SLE if **4 or more** of the following criteria are present, either serially or simultaneously, at any time during observation.

1. Malar Rash

Fixed erythema, flat or raised, over the malar eminences, typically sparing the nasolabial folds.

2. Discoid Rash

Erythematous, raised patches with adherent keratotic scaling and follicular plugging; older lesions may show atrophic scarring.

3. Photosensitivity

Skin rash triggered by an abnormal reaction to sunlight, as documented by patient history or physician observation.

4. Oral Ulcers

Oral or nasopharyngeal ulceration, usually painless, observed by a physician.

5. Arthritis

Non-erosive arthritis involving two or more peripheral joints, characterized by tenderness, swelling, or effusion.

6. *Serositis*

- **Pleuritis:** Convincing history of pleuritic pain, rub heard by a physician, or evidence of pleural effusion.
- **Pericarditis:** Documented by electrocardiogram, pericardial rub, or evidence of pericardial effusion.

7. *Renal Disorder*

- **Persistent proteinuria:** >0.5 g/day or >3+ on dipstick if quantitation not done.
- **Cellular casts:** May include red blood cell, hemoglobin, granular, tubular, or mixed casts.

8. *Neurologic Disorder*

- **Seizures:** Occurring in the absence of offending drugs or known metabolic disturbances (e.g., uremia, ketoacidosis, electrolyte imbalance).
- **Psychosis:** Occurring in the absence of offending drugs or known metabolic disturbances.

9. *Hematologic Disorder*

- **Hemolytic anemia:** With reticulocytosis.
- **Leukopenia:** <4,000/ μ L on two or more occasions.
- **Lymphopenia:** <1,500/ μ L on two or more occasions.
- **Thrombocytopenia:** <100,000/ μ L, in the absence of offending drugs.

10. *Immunologic Disorder*

- **Anti-dsDNA antibodies:** Abnormal titer against native DNA.
- **Anti-Sm antibodies:** Presence of antibody to Sm nuclear antigen.
- **Anti-phospholipid antibodies:** Demonstrated by:
 - Abnormal IgG or IgM anticardiolipin antibody levels,

- Positive test for lupus anticoagulant, or
- False-positive serologic test for syphilis (positive for ≥ 6 months, confirmed by *Treponema pallidum* immobilization or fluorescent treponemal antibody absorption test).

11. Anti-Nuclear Antibody (ANA)

An abnormal titer of ANA by immunofluorescence or equivalent assay, at any time, in the absence of drugs known to cause drug-induced lupus.

Source: Tan EM et al. The 1982 revised criteria for the classification of systemic lupus erythematosus. Arthritis Rheum 25:1271, 1982. Copyright 1982, with permission of the American College of Rheumatology.

Immunopathogenesis of SLE

SLE is associated with the presence of autoreactive B cells in the bone marrow that express high-affinity receptors for self-DNA.

Role of Androgens

Androgens promote a T helper 1 (Th1)-dominant immune response. Both males and females with SLE exhibit reduced levels of testosterone, dihydrotestosterone, dehydroepiandrosterone (DHEA), and DHEA sulfate. Furthermore, androgen-deficient males, such as those with Klinefelter syndrome (47,XXY), have an increased risk of Th2-mediated autoimmune diseases, including SLE.